

Case 5 - Multiple Regression in Biology & Analytics

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1. Introduction

This case study examines how environmental conditions are associated with ecosystem stress using the **Forest Fires** dataset. The response variable is **burned area (area)**, which is used here as a proxy for environmental impact. The analysis applies multiple regression, diagnostic testing, model selection, and validation.

2. Load Packages

```
library(dplyr)
library(tidyr)
library(tidyverse)
library(car)
library(MASS)
library(caret)
library(lmtest)
library(GGally)
```

3. Load Data

```
# Update the path if your file is stored somewhere else
fires <- read.csv("~/Library/CloudStorage/OneDrive-UAEUniversity/Uni/STAT 482/Case 5/forest+fires/forestfires.csv")

# Quick inspection
dim(fires)
```

```
## [1] 517  13
```

```
str(fires)
```

```
## 'data.frame':    517 obs. of  13 variables:
## $ X      : int  7 7 7 8 8 8 8 8 8 7 ...
## $ Y      : int  5 4 4 6 6 6 6 6 6 5 ...
## $ month: chr   "mar" "oct" "oct" "mar" ...
```

```
## $ day : chr "fri" "tue" "sat" "fri" ...
## $ FFMC : num 86.2 90.6 90.6 91.7 89.3 92.3 92.3 91.5 91 92.5 ...
## $ DMC : num 26.2 35.4 43.7 33.3 51.3 ...
## $ DC : num 94.3 669.1 686.9 77.5 102.2 ...
## $ ISI : num 5.1 6.7 6.7 9 9.6 14.7 8.5 10.7 7 7.1 ...
## $ temp : num 8.2 18 14.6 8.3 11.4 22.2 24.1 8 13.1 22.8 ...
## $ RH : int 51 33 33 97 99 29 27 86 63 40 ...
## $ wind : num 6.7 0.9 1.3 4 1.8 5.4 3.1 2.2 5.4 4 ...
## $ rain : num 0 0 0 0.2 0 0 0 0 0 0 ...
## $ area : num 0 0 0 0 0 0 0 0 0 0 ...
```

```
summary(fires)
```

```
##           X           Y           month           day
## Min.      :1.000   Min.      :2.0   Length:517      Length:517
## 1st Qu.:3.000   1st Qu.:4.0   Class :character  Class :character
## Median :4.000   Median :4.0   Mode  :character  Mode  :character
## Mean      :4.669   Mean      :4.3
## 3rd Qu.:7.000   3rd Qu.:5.0
## Max.      :9.000   Max.      :9.0
##           FFMC           DMC           DC           ISI
## Min.      :18.70   Min.      : 1.1   Min.      : 7.9   Min.      : 0.000
## 1st Qu.:90.20   1st Qu.: 68.6   1st Qu.:437.7   1st Qu.: 6.500
## Median :91.60   Median :108.3   Median :664.2   Median : 8.400
## Mean      :90.64   Mean      :110.9   Mean      :547.9   Mean      : 9.022
## 3rd Qu.:92.90   3rd Qu.:142.4   3rd Qu.:713.9   3rd Qu.:10.800
## Max.      :96.20   Max.      :291.3   Max.      :860.6   Max.      :56.100
##           temp           RH           wind           rain
## Min.      : 2.20   Min.      : 15.00   Min.      :0.400   Min.      :0.00000
## 1st Qu.:15.50   1st Qu.: 33.00   1st Qu.:2.700   1st Qu.:0.00000
## Median :19.30   Median : 42.00   Median :4.000   Median :0.00000
## Mean      :18.89   Mean      : 44.29   Mean      :4.018   Mean      :0.02166
## 3rd Qu.:22.80   3rd Qu.: 53.00   3rd Qu.:4.900   3rd Qu.:0.00000
## Max.      :33.30   Max.      :100.00   Max.      :9.400   Max.      :6.40000
##           area
## Min.      : 0.00
## 1st Qu.: 0.00
## Median : 0.52
## Mean      : 12.85
## 3rd Qu.: 6.57
## Max.      :1090.84
```

```
head(fires)
```

```
##   X Y month day FFMC DMC DC ISI temp RH wind rain area
## 1 7 5 mar fri 86.2 26.2 94.3 5.1 8.2 51 6.7 0.0 0
## 2 7 4 oct tue 90.6 35.4 669.1 6.7 18.0 33 0.9 0.0 0
## 3 7 4 oct sat 90.6 43.7 686.9 6.7 14.6 33 1.3 0.0 0
## 4 8 6 mar fri 91.7 33.3 77.5 9.0 8.3 97 4.0 0.2 0
## 5 8 6 mar sun 89.3 51.3 102.2 9.6 11.4 99 1.8 0.0 0
## 6 8 6 aug sun 92.3 85.3 488.0 14.7 22.2 29 5.4 0.0 0
```

4. Data Understanding and Cleaning

4.1 Check missing values

```
colSums(is.na(fires))
```

```
##      X      Y month    day  FPMC   DMC   DC   ISI  temp   RH  wind  rain  area
##      0      0      0      0      0      0      0      0      0      0      0      0      0
```

4.2 Convert categorical variables

```
fires$month <- factor(fires$month,
                      levels = c("jan", "feb", "mar", "apr", "may", "jun",
                                   "jul", "aug", "sep", "oct", "nov", "dec"),
                      ordered = TRUE)

fires$day <- factor(fires$day,
                   levels = c("mon", "tue", "wed", "thu", "fri", "sat", "sun"),
                   ordered = TRUE)

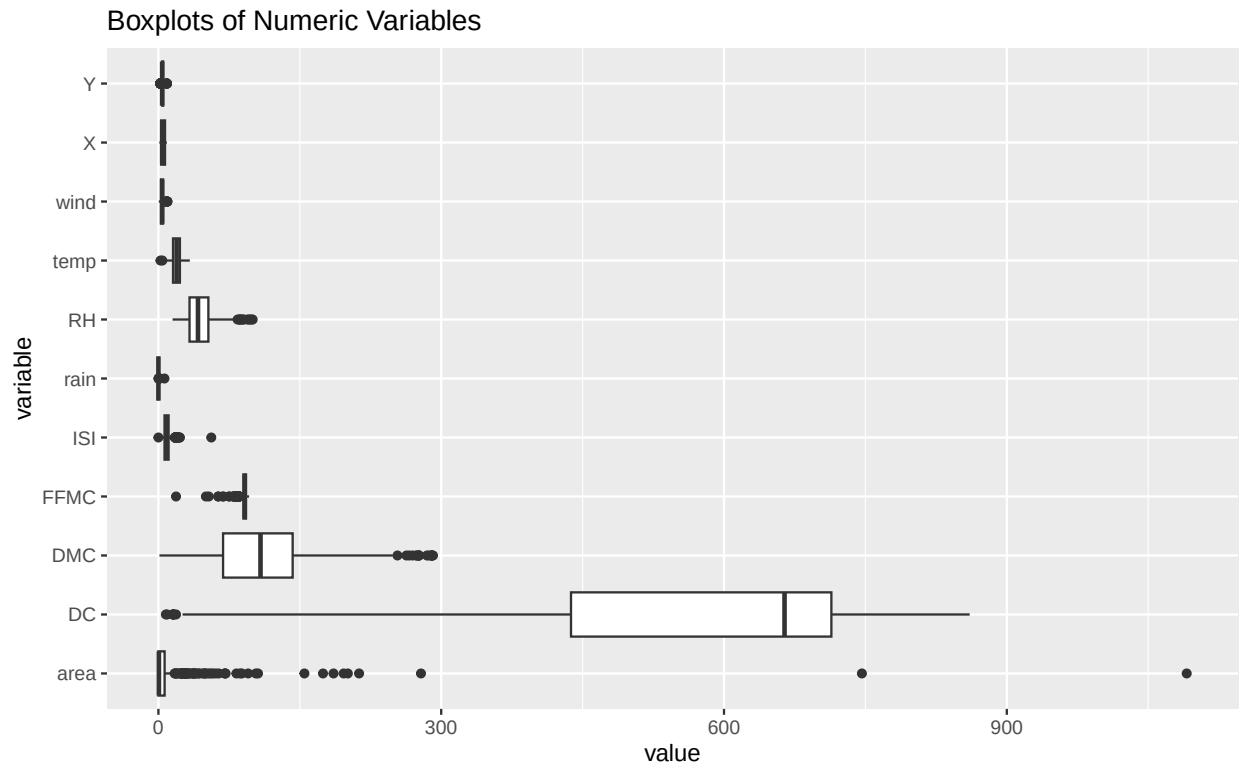
str(fires)
```

```
## 'data.frame':    517 obs. of  13 variables:
## $ X      : int  7 7 7 8 8 8 8 8 8 7 ...
## $ Y      : int  5 4 4 6 6 6 6 6 6 5 ...
## $ month: Ord.factor w/ 12 levels "jan"<"feb"<"mar"<...: 3 10 10 3 3 8 8 8 9 9 ...
## $ day   : Ord.factor w/ 7 levels "mon"<"tue"<"wed"<...: 5 2 6 5 7 7 1 1 2 6 ...
## $ FPMC  : num  86.2 90.6 90.6 91.7 89.3 92.3 92.3 91.5 91 92.5 ...
## $ DMC   : num  26.2 35.4 43.7 33.3 51.3 ...
## $ DC    : num  94.3 669.1 686.9 77.5 102.2 ...
## $ ISI   : num  5.1 6.7 6.7 9 9.6 14.7 8.5 10.7 7 7.1 ...
## $ temp  : num  8.2 18 14.6 8.3 11.4 22.2 24.1 8 13.1 22.8 ...
## $ RH    : int  51 33 33 97 99 29 27 86 63 40 ...
## $ wind  : num  6.7 0.9 1.3 4 1.8 5.4 3.1 2.2 5.4 4 ...
## $ rain  : num  0 0 0 0.2 0 0 0 0 0 0 ...
## $ area  : num  0 0 0 0 0 0 0 0 0 0 ...
```

4.3 Check outliers

```
numeric_vars <- fires %>%
  dplyr::select(where(is.numeric))

numeric_vars %>%
  pivot_longer(cols = everything(), names_to = "variable", values_to = "value") %>%
  ggplot(aes(x = variable, y = value)) +
  geom_boxplot() +
  coord_flip() +
  labs(title = "Boxplots of Numeric Variables")
```

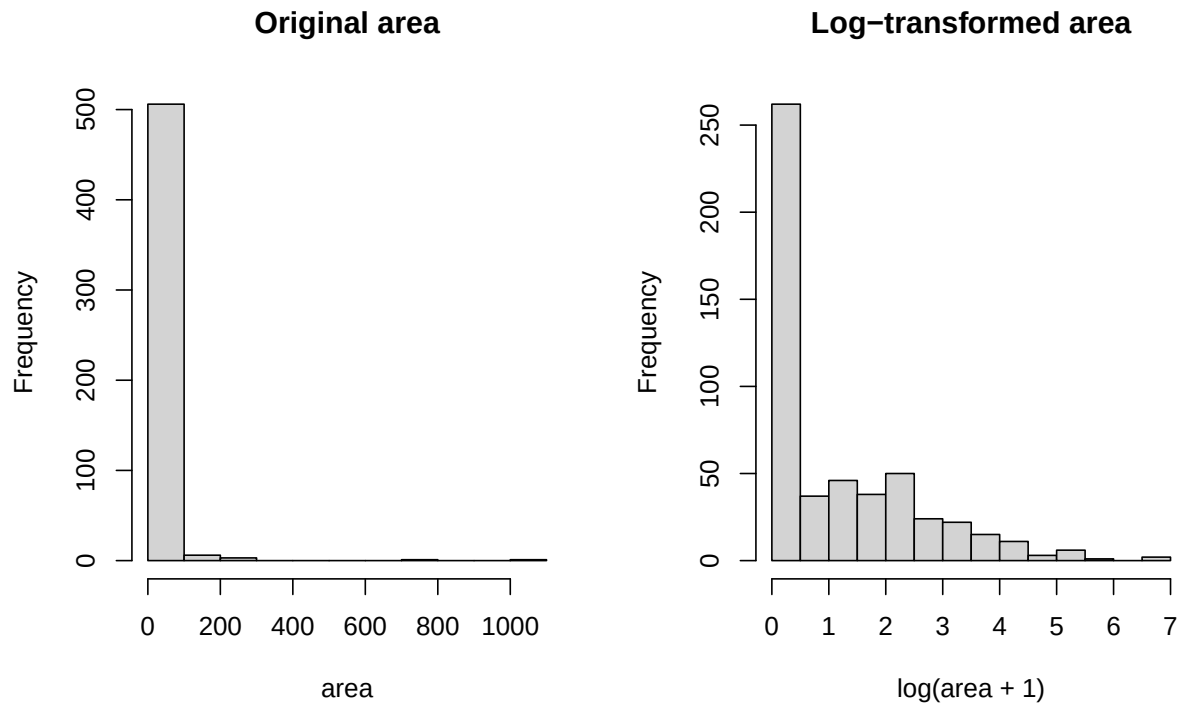


4.4 Transform the response variable

The variable `area` is heavily right-skewed, so we use $\log(\text{area} + 1)$.

```
fires$log_area <- log(fires$area + 1)

par(mfrow = c(1, 2))
hist(fires$area, main = "Original area", xlab = "area")
hist(fires$log_area, main = "Log-transformed area", xlab = "log(area + 1)")
```



```
par(mfrow = c(1, 1))
```

5. Exploratory Data Analysis

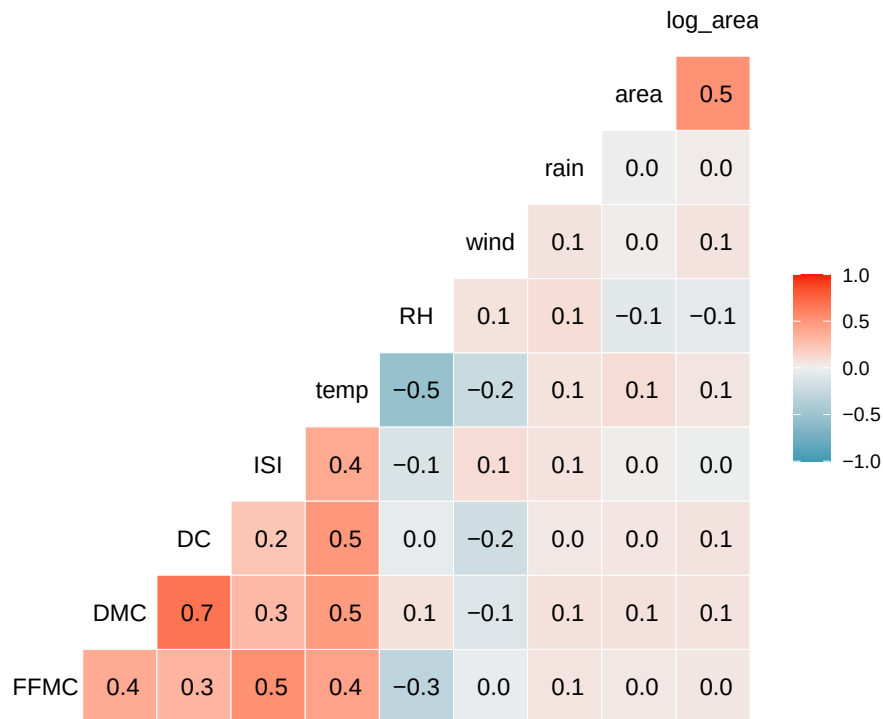
5.1 Correlation among numeric variables

```
cor_data <- fires %>%
  dplyr::select(FFMC, DMC, DC, ISI, temp, RH, wind, rain, area, log_area)

cor_matrix <- cor(cor_data, use = "complete.obs")
round(cor_matrix, 2)
```

	FFMC	DMC	DC	ISI	temp	RH	wind	rain	area	log_area
## FFMC	1.00	0.38	0.33	0.53	0.43	-0.30	-0.03	0.06	0.04	0.05
## DMC	0.38	1.00	0.68	0.31	0.47	0.07	-0.11	0.07	0.07	0.07
## DC	0.33	0.68	1.00	0.23	0.50	-0.04	-0.20	0.04	0.05	0.07
## ISI	0.53	0.31	0.23	1.00	0.39	-0.13	0.11	0.07	0.01	-0.01
## temp	0.43	0.47	0.50	0.39	1.00	-0.53	-0.23	0.07	0.10	0.05
## RH	-0.30	0.07	-0.04	-0.13	-0.53	1.00	0.07	0.10	-0.08	-0.05
## wind	-0.03	-0.11	-0.20	0.11	-0.23	0.07	1.00	0.06	0.01	0.07
## rain	0.06	0.07	0.04	0.07	0.07	0.10	0.06	1.00	-0.01	0.02
## area	0.04	0.07	0.05	0.01	0.10	-0.08	0.01	-0.01	1.00	0.52
## log_area	0.05	0.07	0.07	-0.01	0.05	-0.05	0.07	0.02	0.52	1.00

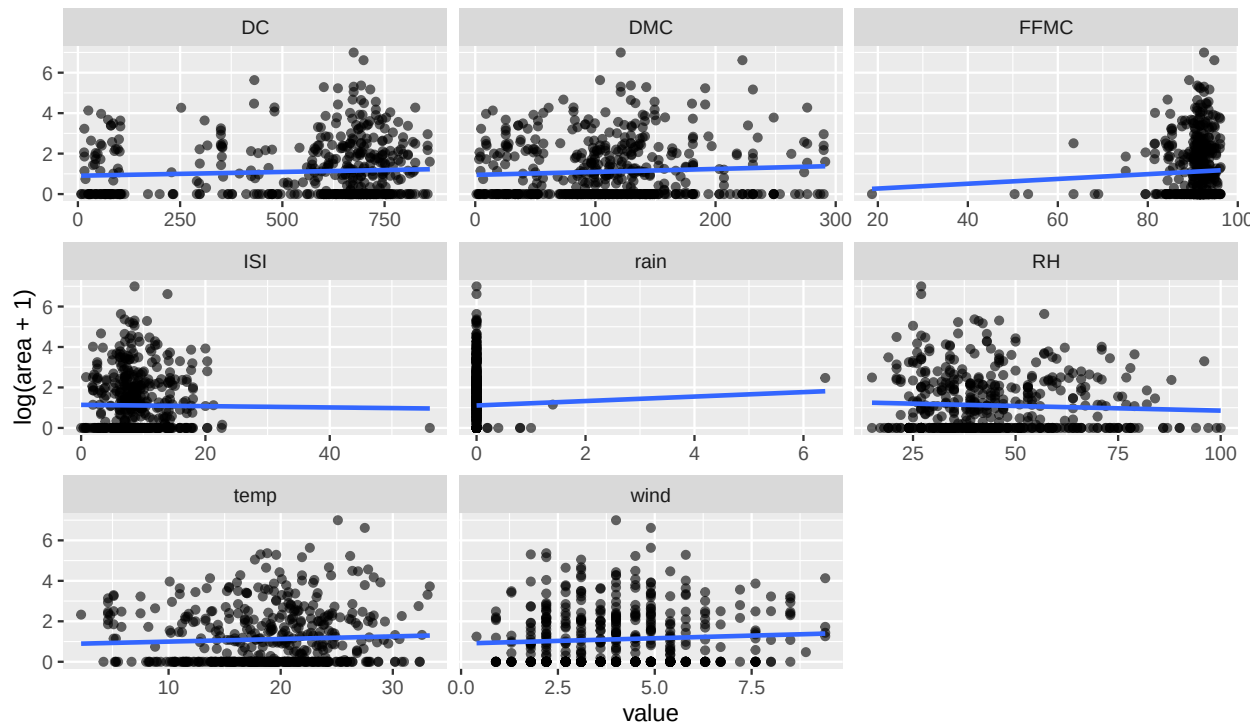
```
GGally::ggcorr(cor_data, label = TRUE)
```



5.2 Scatterplots against the response

```
fires %>%
  pivot_longer(cols = c(FFMC, DMC, DC, ISI, temp, RH, wind, rain),
    names_to = "predictor",
    values_to = "value") %>%
  ggplot(aes(x = value, y = log_area)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ predictor, scales = "free_x") +
  labs(title = "Predictors vs Log Burned Area", y = "log(area + 1)")
```

Predictors vs Log Burned Area



6. Model Specification

6.1 Full multiple regression model

```
full_model <- lm(log_area ~ FFMC + DMC + DC + ISI + temp + RH + wind + rain,
                 data = fires)

summary(full_model)
```

```
##
## Call:
## lm(formula = log_area ~ FFMC + DMC + DC + ISI + temp + RH + wind +
##     rain, data = fires)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.5203 -1.1129 -0.6158  0.8787  5.7121
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.2224140   1.3604350   0.163   0.870
## FFMC         0.0077082   0.0144884   0.532   0.595
## DMC          0.0011915   0.0014642   0.814   0.416
## DC           0.0002737   0.0003570   0.767   0.444
## ISI        -0.0239494   0.0169248  -1.415   0.158
```

```
## temp      0.0024618  0.0172593  0.143  0.887
## RH        -0.0051729  0.0051889 -0.997  0.319
## wind      0.0757669  0.0366155  2.069  0.039 *
## rain      0.0965122  0.2121461  0.455  0.649
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.395 on 508 degrees of freedom
## Multiple R-squared:  0.01988,    Adjusted R-squared:  0.004446
## F-statistic: 1.288 on 8 and 508 DF,  p-value: 0.2472
```

6.2 Optional model with categorical controls

```
model_time <- lm(log_area ~ FPMC + DMC + DC + ISI + temp + RH + wind + rain + month + day,
                 data = fires)

summary(model_time)
```

```
##
## Call:
## lm(formula = log_area ~ FPMC + DMC + DC + ISI + temp + RH + wind +
##     rain + month + day, data = fires)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.9115 -1.0352 -0.5310  0.7901  5.2962
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.046193   1.617344  -0.029   0.9772
## FPMC         0.007537   0.016660   0.452   0.6512
## DMC          0.004073   0.001863   2.187   0.0292 *
## DC          -0.001876   0.001258  -1.491   0.1367
## ISI         -0.013408   0.017982  -0.746   0.4562
## temp         0.037199   0.022252   1.672   0.0952 .
## RH           0.001599   0.006212   0.257   0.7970
## wind         0.061519   0.038424   1.601   0.1100
## rain         0.051398   0.214711   0.239   0.8109
## month.L      1.206085   1.040426   1.159   0.2469
## month.Q      0.353297   0.902257   0.392   0.6955
## month.C      0.710602   0.845895   0.840   0.4013
## month^4      0.631993   0.677071   0.933   0.3511
## month^5      1.105487   0.888276   1.245   0.2139
## month^6      1.190056   0.778713   1.528   0.1271
## month^7      1.354255   0.699810   1.935   0.0535 .
## month^8     -0.093554   0.543601  -0.172   0.8634
## month^9      0.233368   0.388468   0.601   0.5483
## month^10     0.255285   0.545244   0.468   0.6398
## month^11    -0.431065   0.470931  -0.915   0.3605
## day.L       -0.013194   0.160342  -0.082   0.9345
## day.Q        0.095974   0.165602   0.580   0.5625
## day.C        0.100508   0.166607   0.603   0.5466
```

```
## day^4      -0.217867   0.168655  -1.292   0.1970
## day^5      -0.108609   0.170895  -0.636   0.5254
## day^6      -0.069246   0.176305  -0.393   0.6947
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.384 on 491 degrees of freedom
## Multiple R-squared:  0.06842,    Adjusted R-squared:  0.02098
## F-statistic: 1.442 on 25 and 491 DF,  p-value: 0.07785
```

6.3 Optional interaction example

```
model_interaction <- lm(log_area ~ FFMC + DMC + DC + ISI + temp * RH + wind + rain,
                        data = fires)

summary(model_interaction)
```

```
##
## Call:
## lm(formula = log_area ~ FFMC + DMC + DC + ISI + temp * RH + wind +
##     rain, data = fires)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.5795 -1.1004 -0.6119  0.8728  5.6610
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.3390215   1.5469937  -0.219   0.8266
## FFMC         0.0103761   0.0149097   0.696   0.4868
## DMC          0.0011780   0.0014649   0.804   0.4217
## DC           0.0003040   0.0003593   0.846   0.3980
## ISI          -0.0259566   0.0171347  -1.515   0.1304
## temp         0.0215696   0.0304078   0.709   0.4784
## RH           0.0023608   0.0111507   0.212   0.8324
## wind         0.0797857   0.0370069   2.156   0.0316 *
## rain         0.1255872   0.2156236   0.582   0.5605
## temp:RH      -0.0004941   0.0006473  -0.763   0.4456
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.396 on 507 degrees of freedom
## Multiple R-squared:  0.02101,    Adjusted R-squared:  0.003628
## F-statistic: 1.209 on 9 and 507 DF,  p-value: 0.287
```

7. Multicollinearity Check

```
vif(full_model)
```

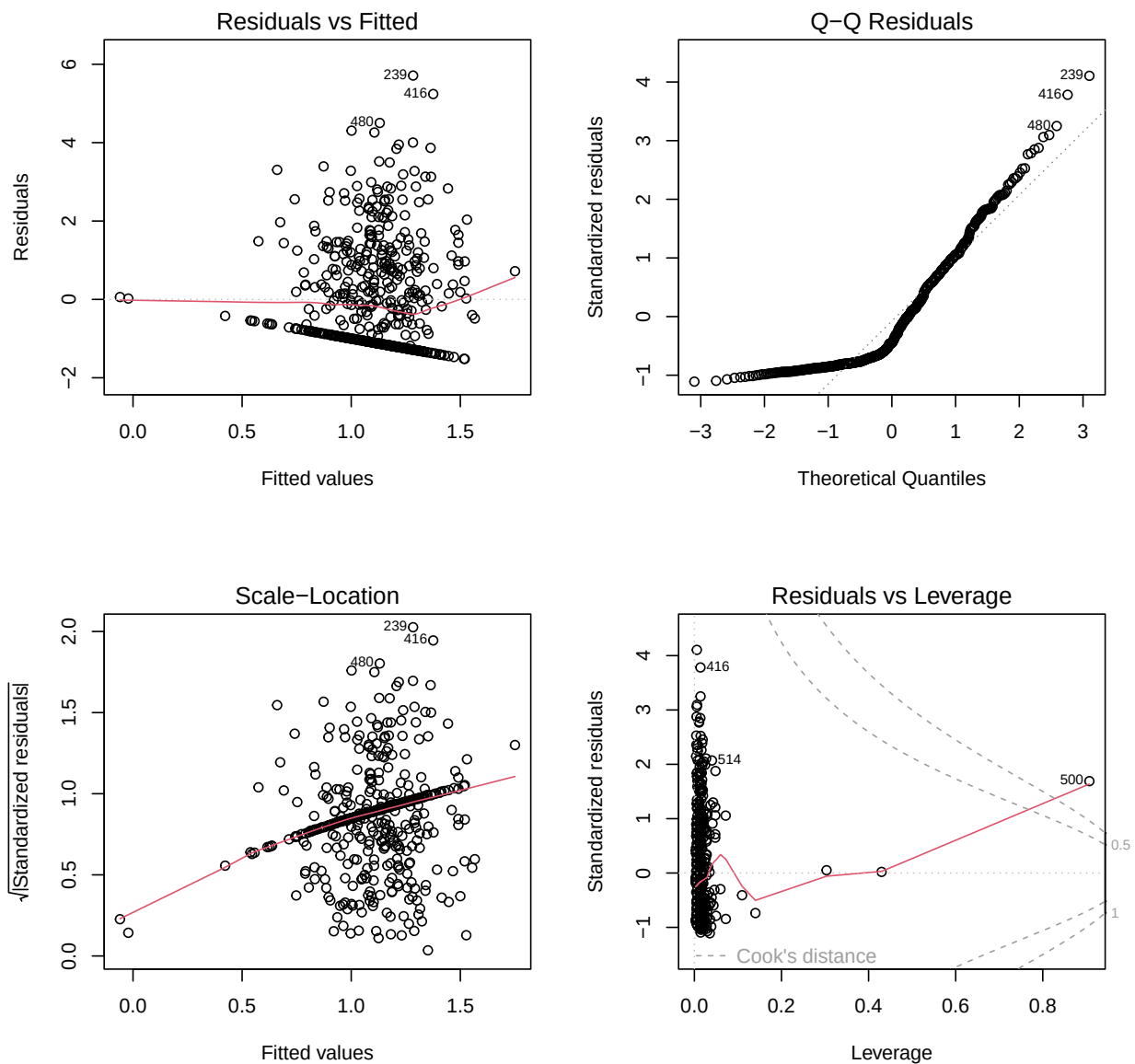
```
##      FPMC      DMC      DC      ISI      temp      RH      wind      rain
## 1.695255 2.330688 2.078205 1.578258 2.661897 1.899989 1.140610 1.044801
```

Interpretation guideline: - VIF < 5: usually acceptable - VIF 5-10: moderate concern - VIF > 10: serious multicollinearity concern

8. Model Diagnostics

8.1 Residual plots

```
par(mfrow = c(2, 2))
plot(full_model)
```



```
par(mfrow = c(1, 1))
```

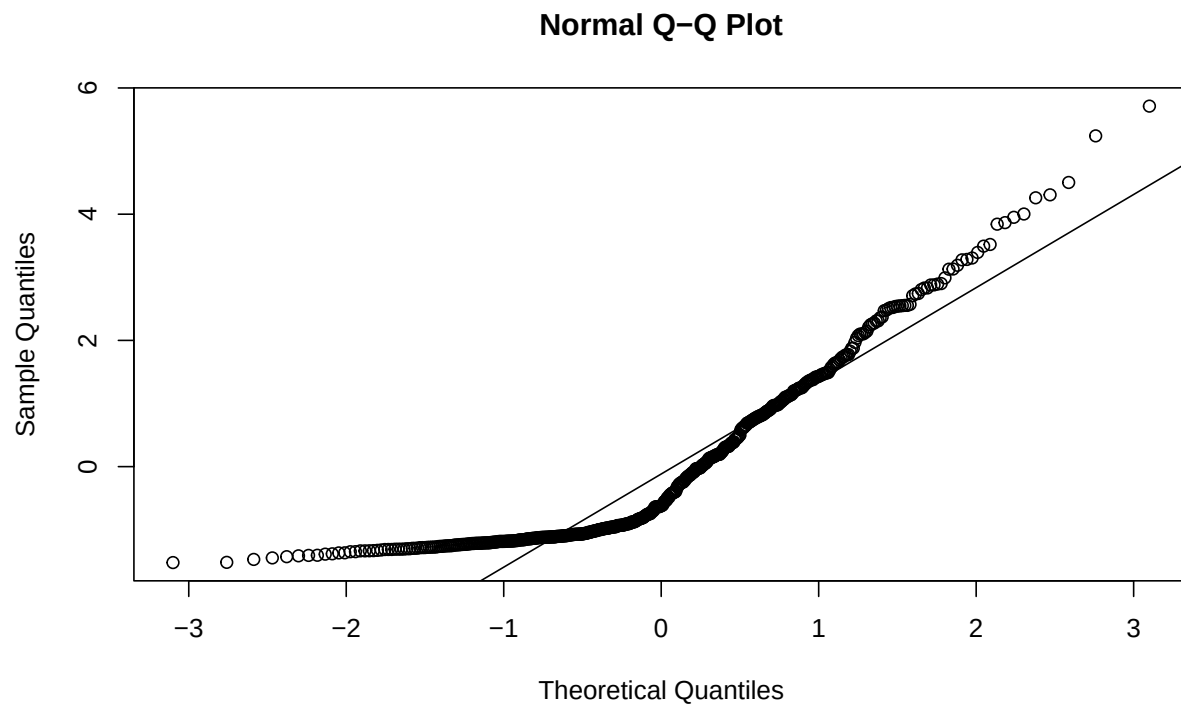
8.2 Breusch-Pagan test for homoscedasticity

```
bptest(full_model)
```

```
##
## studentized Breusch-Pagan test
##
## data: full_model
## BP = 11.27, df = 8, p-value = 0.1868
```

8.3 Normality check for residuals

```
{  
  qqnorm(resid(full_model))  
  qqline(resid(full_model))  
}
```



```
shapiro.test(resid(full_model))
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  resid(full_model)  
## W = 0.85423, p-value < 2.2e-16
```

9. Model Selection

9.1 Compare candidate models

```
model_comparison <- tibble(  
  Model = c("Full Model", "Time Controls", "Interaction Model"),  
  Adjusted_R2 = c(summary(full_model)$adj.r.squared,  
                  summary(model_time)$adj.r.squared,
```

```

summary(model_interaction)$adj.r.squared),
AIC = c(AIC(full_model), AIC(model_time), AIC(model_interaction)),
BIC = c(BIC(full_model), BIC(model_time), BIC(model_interaction))
)

model_comparison

```

```

## # A tibble: 3 x 4
##   Model      Adjusted_R2    AIC    BIC
##   <chr>          <dbl> <dbl> <dbl>
## 1 Full Model      0.00445 1823. 1865.
## 2 Time Controls   0.0210  1830. 1945.
## 3 Interaction Model 0.00363 1824. 1871.

```

9.2 Stepwise regression

```

step_model <- stepAIC(full_model, direction = "both", trace = FALSE)
summary(step_model)

```

```

##
## Call:
## lm(formula = log_area ~ DMC + RH + wind, data = fires)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.5093 -1.1071 -0.6537  0.9216  5.7712
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.9129363  0.2433065   3.752 0.000195 ***
## DMC          0.0017552  0.0009657   1.817 0.069731 .
## RH          -0.0055830  0.0037785  -1.478 0.140138
## wind         0.0624134  0.0345109   1.809 0.071112 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.392 on 513 degrees of freedom
## Multiple R-squared:  0.01425,    Adjusted R-squared:  0.008485
## F-statistic: 2.472 on 3 and 513 DF,  p-value: 0.06101

```

```
AIC(step_model)
```

```
## [1] 1815.518
```

```
BIC(step_model)
```

```
## [1] 1836.758
```

10. Validation

10.1 Train/Test split

```
set.seed(202031137)
train_index <- createDataPartition(fires$log_area, p = 0.80, list = FALSE)
train_data <- fires[train_index, ]
test_data <- fires[-train_index, ]

dim(train_data)
```

```
## [1] 416 14
```

```
dim(test_data)
```

```
## [1] 101 14
```

10.2 Fit selected model on training data

```
train_model <- lm(log_area ~ FPMC + DMC + DC + ISI + temp + RH + wind + rain,
                  data = train_data)
summary(train_model)
```

```
##
## Call:
## lm(formula = log_area ~ FPMC + DMC + DC + ISI + temp + RH + wind +
##     rain, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.5831 -1.0618 -0.5655  0.9072  5.3127
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.2581721  1.4067582   0.184   0.8545
## FPMC         0.0045570  0.0148461   0.307   0.7590
## DMC          0.0004967  0.0015905   0.312   0.7550
## DC           0.0006025  0.0003765   1.600   0.1103
## ISI          -0.0298885  0.0174809  -1.710   0.0881 .
## temp         0.0062346  0.0186917   0.334   0.7389
## RH           -0.0029577  0.0055909  -0.529   0.5971
## wind         0.0828639  0.0395539   2.095   0.0368 *
## rain        -0.8911564  0.7521886  -1.185   0.2368
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.349 on 407 degrees of freedom
## Multiple R-squared:  0.03022,    Adjusted R-squared:  0.01116
## F-statistic: 1.585 on 8 and 407 DF,  p-value: 0.1271
```

10.3 Predict on test data

```
pred_log <- predict(train_model, newdata = test_data)

rmse_value <- RMSE(pred_log, test_data$log_area)
mae_value <- MAE(pred_log, test_data$log_area)

validation_results <- tibble(
  Metric = c("RMSE", "MAE"),
  Value = c(rmse_value, mae_value)
)

validation_results
```

```
## # A tibble: 2 x 2
##   Metric Value
##   <chr>   <dbl>
## 1 RMSE    1.71
## 2 MAE     1.30
```

11. Interpretation Template

Use this section after running the model:

- **Coefficient signs:** Explain whether each significant predictor increases or decreases expected $\log(\text{area} + 1)$.
- **Magnitude:** For each 1-unit increase in a predictor, the response changes by the corresponding coefficient, holding all other variables constant.
- **Most influential factors:** Focus on predictors with statistically significant p-values and practical relevance.
- **Biological meaning:** Relate factors such as temperature, humidity, rainfall, and drought indicators to ecosystem stress.
- **Business implications:** Discuss implications for agricultural planning, wildfire risk mitigation, and controlled ecosystem management.

12. Conclusion Template

This analysis applies multiple regression to evaluate how environmental variables influence ecosystem impact. After checking assumptions, selecting the best model, and validating performance, the final model can be used to identify the most influential environmental drivers and support data-driven decision-making.

13. Optional: Save Key Outputs

```
write.csv(model_comparison, "model_comparison.csv", row.names = FALSE)
write.csv(validation_results, "validation_results.csv", row.names = FALSE)
```